14	**D**	The pendulum performs simple harmonic motion at small angles. You may see the pendulum swing as a part of a non-uniform vertical circular motion. Let angular displacement be θ. Thus, linear displacement $x \approx r\theta = (1.0)\theta = \theta$ At angular displacement is 0.050 rad, horizontal amplitude x_0 = 0.050 m. At angular displacement is 0.030 rad, horizontal displacement x = 0.030 m. Using the equation given in the formula list, linear velocity, $v = \pm\omega\sqrt{x_0^2 - x^2}$ $= \pm\frac{2\pi}{T}\sqrt{x_0^2 - x^2}$ $= \pm\frac{2\pi}{2.0}\sqrt{0.050^2 - 0.030^2}$ $= \pm 0.13$ m s^{-1} $\because v = r\omega = (1.0)\omega = \omega$ angular speed, $\omega = v = 0.13$ rads^{-1}
15	**B**	Using Malus' law, the intensity of transmitted light (I) through polarisers is proportional to